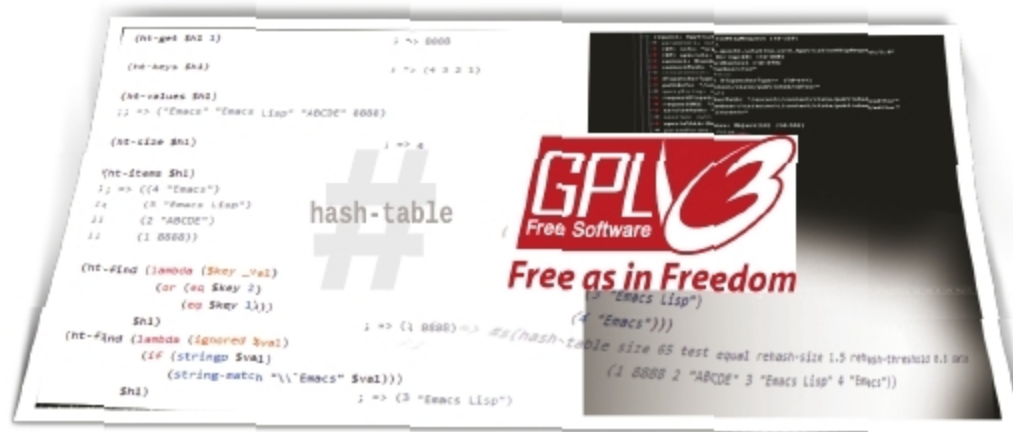


The Emacs Series

ht.el: The Hash Table Library for Emacs



In this article, we explore the various hash table functions and macros provided by the ht.el library.

The ht.el hash table library for Emacs has been written by Wilfred Hughes. The latest tagged release is version 2.2 and the software is released under the GNU General Public License v3. The source code is available at <https://github.com/Wilfred/ht.el>. It provides a comprehensive list of hash table operations and a very consistent API. For example, any mutation function will always return nil.

Installation

The Milkypostman's Emacs Lisp Package Archive (MELPA) and Marmalade repositories have ht.el available for installation. You can add the following command to your Emacs *init.el* configuration file:

```
(require 'package)
(add-to-list 'package-archives '("melpa" . "https://melpa.
org/packages/") t)
```

You can then run *M-x package <RET> ht <RET>* to install the ht.el library. If you are using Cask, then you simply add the following code to your Cask file:

```
(depends-on "ht")
```

You will need the *ht* library in your Emacs environment before using the API functions.

```
(require 'ht)
```

Usage

Let us now explore the various API functions provided by the ht.el library. The *ht-create* function will return a hash table that can be assigned to a hash table variable. You can also verify that the variable is a hash table using the *type-of* function as shown below:

```
(let ((greetings (ht-create)))
  (type-of greetings))
```

```
hash-table
```

You can add an item to the hash table using the *ht-set!* function, which takes the hash table, key and value as arguments. The entries in the hash table can be listed using the *ht-items* function as illustrated below:

```
(ht-set! hash-table key value) ;; Syntax
(ht-items hash-table)          ;; Syntax
```

```
(let ((greetings (ht-create)))
  (ht-set! greetings "Adam" "Hello Adam!")
  (ht-items greetings))

(("Adam" "Hello Adam!"))
```

The keys present in a hash table can be retrieved using the *ht-keys* function, while the values in a hash table can be obtained using the *ht-values* function, as shown in the following examples:

```
(ht-keys hash-table) ;; Syntax
(ht-values hash-table) ;; Syntax
```

```
(let ((greetings (ht-create)))
  (ht-set! greetings "Adam" "Hello Adam!")
  (ht-keys greetings))
```

```
("Adam")
```

```
(let ((greetings (ht-create)))
  (ht-set! greetings "Adam" "Hello Adam!")
  (ht-values greetings))
```

```
("Hello Adam!")
```

The "ht-clear!" function can be used to clear all the items in a hash-table. For example:

```
(ht-clear! hash-table) ;; Syntax
```